

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Nuclear Geophysics Lab L-T-P: 0-0-3 Credits:2

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Determination of the efficiency of nuclear counters, study of nuclear counting statistics, estimation of resolving time of counters, uranium percentage by comparison, determination of uranium and thorium in a geological sample by simultaneous beta and gamma measurements, absorption of alpha, beta particles and gamma rays, in various media viz., lead, aluminum and light elements, determination of linear and mass absorption coefficients, variation of radioactivity with distance and source configuration, concept of blind zone, calibration and measurements using portable gamma ray spectrometers for geophysical applications, radon emanation from soils and rocks using active and passive devices, studies on radon migration in relation to fractures and faulting in rocks.

Reference Books:

- 1) Radiation Detection and Measurements by G.F. Knoll
- 2) Nuclear Measurement Techniques by K.S. Ram

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Determination of the efficiency of nuclear counters, study of nuclear counting statistics	3
2.	Estimation of resolving time of counters, uranium percentage by comparison	2
3.	Determination of uranium and thorium in a geological sample by simultaneous beta and gamma measurements, absorption of alpha, beta particles and gamma rays, in various media viz., lead, aluminum and light elements	4
4.	Determination of linear and mass absorption coefficients, variation of radioactivity with distance and source configuration	3

5	Concept of blind zone, calibration and measurements using portable gamma ray spectrometers for geophysical applications, radon emanation from soils and rocks using active and passive devices	4
6	Studies on radon migration in relation to fractures and faulting in rocks.	2
Total number of hours		18